Study Guide MS-102 — Microsoft 365 Administrator

> 🌐 **Language:** English | Nederlands

Duration: 8 weeks · **Lab preset:** Standard (DC01 · MGMT01 · W11-01 · W11-02) **MS Learn path:** Microsoft 365 Administrator **Exam weight:**

Domain	Weight
Deploy and manage a Microsoft 365 tenant	25–30%
Implement and manage Microsoft Entra identity and access	25–30%
Manage security and threats by using Microsoft Defender XDR	30–35%
Manage compliance by using Microsoft Purview	10–15%

> **Updated:** Skills measured as of April 28, 2026 (study guide). Domain weights and structure unchanged from November 10, 2025. Minor updates to users/groups management, role groups, secure access, Defender for Endpoint, and DLP scope (now explicitly includes Power BI and Microsoft 365 Copilot). ⚠️ **Note for candidates taking the exam before April 28, 2026:** The current live exam (exam page) uses different domain weights — Tenant 10–15%, Defender XDR 35–40%, Purview 15–20%. The study guide weights above apply as of April 28, 2026. Always check the official exam page before scheduling.

> **Prerequisite:** Microsoft 365 E5 developer tenant (via MSDN/M365 developer program) Sign up at: developer.microsoft.com/microsoft-365/dev-program **Tenant note:** MS-102 uses the local VMs and the shared dev tenant together. Whenever a topic moves into the Microsoft 365 admin center, Defender, Purview, or Entra portal, switch to the tenant explicitly.

How to use this study guide

- Start each week with the learning objectives and MS Learn modules; MS-102 rewards understanding how services connect, not just memorising isolated facts.
- Run the lab exercises in both the tenant and the listed VMs so you can see the relationship between cloud administration and hybrid reality.
- Complete the knowledge check only after theory and hands-on work, and explicitly mark the questions you only answered "roughly right".

- When working with Dutch-speaking and English-speaking colleagues, keep the same core terms active in both languages, for example *retention policy / retentiebeleid* and *role assignment / roltoewijzing*.

Lab coverage and expectations

- **Strong SSW-Lab coverage:** tenant administration, Entra Connect, baseline admin roles, licensing, Conditional Access, Defender for Endpoint, identity hygiene, and part of Purview administration.
- **Partial coverage:** Defender for Office 365, Defender for Cloud Apps, Insider Risk, advanced Purview workflows, and some compliance features depend heavily on licensing and tenant availability.
- **Cloud-only or only partly reproducible:** Microsoft 365 Backup experience, production mail-flow scenarios, external domain onboarding, and some cross-workload Defender XDR integrations.
- Treat the lab as your primary practice environment, but plan direct portal exploration or MS Learn review for the marked areas.

How to use the knowledge checks

- Answer first from the perspective of what you would actually do in an admin role.
- Then verify that your answer is not only technically correct, but also aligned with least privilege, governance, and operational reality.
- Revisit weaker questions one day later; MS-102 often tests chain reasoning across multiple portals and services.
- Watch out for near-neighbour terms such as *security group*, *Microsoft 365 group*, *directory role*, and *Azure RBAC role*.

Week 1 — Setting up the Microsoft 365 tenant

> **Exam domain:** Deploy and manage a Microsoft 365 tenant · **Weight:** 25–30%

> **Real-world scenario:** A Sogeti consultant is engaged to onboard a mid-sized professional services firm (350 users) to Microsoft 365. The client owns a custom domain hosted at an external DNS provider and currently runs all user accounts in on-premises Active Directory. The consultant must set up the tenant, verify the domain, connect the directory, and ensure all users receive licenses before the planned cutover date.

Learning Objectives

- [ ] Navigate the Microsoft 365 admin center and identify its key management areas
- [ ] Configure tenant settings including organisation name, region, and time zone
- [ ] Add and verify a custom domain by creating the required DNS records
- [ ] Install and configure Azure AD Connect to synchronise on-premises AD users to Entra ID

- [ ] Assign Microsoft 365 E5 licenses to synchronised users
- [ ] Enable Microsoft 365 Backup and define backup policies for Exchange and OneDrive

MS Learn modules

- Explore your Microsoft 365 cloud environment
- Configure your Microsoft 365 experience
- Manage Microsoft 365 tenants

Key Concepts

Term	Description
Microsoft 365 admin center	Central management portal for Microsoft 365 services, users, licenses, and billing (admin.microsoft.com)
Managed domain	A domain where authentication is handled entirely in Entra ID (cloud-only or password hash sync)
Federated domain	A domain where authentication is redirected to an on-premises identity provider (e.g., ADFS)
Azure AD Connect	Tool that synchronises on-premises Active Directory objects to Microsoft Entra ID
Password Hash Synchronisation (PHS)	Sync method that copies a hash of the user's password hash to Entra ID, allowing cloud-based authentication
Pass-Through Authentication (PTA)	Sync method that validates passwords against on-premises AD in real time without storing any hash in the cloud
Microsoft 365 Backup	Native backup service for Exchange, OneDrive, and SharePoint that enables point-in-time restore, distinct from retention policies
DNS records for custom domain	Includes a TXT verification record, MX record for mail delivery, CNAME records for Autodiscover and other services

Lab exercises (SSW-Lab)

VM	Task
LAB-MGMT01	Open the **Microsoft 365 admin center** (admin.microsoft.com) in Edge
LAB-MGMT01	Configure tenant settings: name, region, time zone
LAB-MGMT01	Add a custom domain (or verify `ssw.lab` equivalent in tenant)
LAB-DC01	Install Azure AD Connect → synchronise `ssw.lab` AD users to Entra ID
LAB-MGMT01	Verify synchronised users in **Entra admin center** → **Users**
LAB-MGMT01	Assign Microsoft 365 E5 licenses to synchronised users

VM	Task
LAB-MGMT01	Enable and configure **Microsoft 365 Backup**: set a backup policy for Exchange and OneDrive

Lab commands

```
# Connect to Microsoft Graph (required for most tenant admin tasks)
Connect-MgGraph -Scopes "User.ReadWrite.All", "Organization.ReadWrite.All"

# Verify domain status after adding custom domain
Get-MgDomain | Select-Object Id, IsVerified, IsDefault

# Trigger a manual delta sync from Entra Connect server
Start-ADSyncSyncCycle -PolicyType Delta

# Assign E5 license to a user (replace with real SKU GUID from Get-
MgSubscribedSku)
Set-MgUserLicense -UserId "user@ssw.lab" -AddLicenses @{SkuId = "<E5-SKU-GUID>"}
-RemoveLicenses @()
```

Knowledge check

1. What is the difference between a *managed domain* and a *federated domain* in Microsoft 365?
2. How does *password hash synchronisation* work versus *pass-through authentication*?
3. Which DNS records are required for a custom domain in Microsoft 365?
4. What is the *Microsoft 365 compliance center* and what is it used for?
5. What is *Microsoft 365 Backup* and how does it differ from retention policies?

▶ Answers

Week 2 — User and group management

> **Exam domain:** Deploy and manage a Microsoft 365 tenant · **Weight:** 25–30%

> **Real-world scenario:** A Sogeti consultant is helping a logistics company (800 users, 12 departments) streamline their Microsoft 365 administration. The IT team currently assigns licenses individually and resets passwords manually. The consultant needs to implement dynamic groups for automatic license assignment, configure SSPR to reduce helpdesk load, and introduce PIM to protect the three Global Administrator accounts that are currently always active.

Learning Objectives

- ☐ Apply the principle of least privilege when assigning Microsoft 365 admin roles
- ☐ Create and manage users in bulk using CSV import and PowerShell
- ☐ Configure dynamic groups in Entra ID using attribute-based membership rules
- ☐ Enable and test Self-Service Password Reset (SSPR) for a defined group
- ☐ Configure Privileged Identity Management (PIM) to make admin roles eligible rather than permanently active
- ☐ Activate a role Just-in-Time via PIM and verify the audit trail

MS Learn modules

- Manage users and groups in Microsoft 365
- Manage admin roles in Microsoft 365
- Manage password policies

Key Concepts

Term	Description
Least privilege	Security principle: assign only the minimum permissions required for a task — no more, no less
Security group	Entra ID group used to grant access to resources; does not provide Microsoft 365 services or email
Microsoft 365 group	Shared workspace with mailbox, calendar, SharePoint site, and Teams channel; backed by Entra ID
Dynamic group	Group whose membership is automatically maintained by Entra ID based on user attribute rules (e.g., department, jobTitle)
Assigned group	Group with manually managed membership; suitable when membership is static or exception-based
Self-Service Password Reset (SSPR)	Allows users to reset their own password without contacting helpdesk, using registered authentication methods
Privileged Identity Management (PIM)	Entra ID feature that provides Just-in-Time privileged access, requiring activation before a role becomes active
Just-in-Time (JIT) access	Pattern in which elevated rights are granted only for a limited time window upon explicit request and approval

Lab exercises (SSW-Lab)

VM	Task
LAB-DC01	Create OU structure: `OU=LAB, OU=Users, DC=ssw, DC=lab`

VM	Task
LAB-DC01	Bulk-create test accounts via CSV + `Import-Csv | New-ADUser`
LAB-MGMT01	Assign roles in M365: create a *Helpdesk Administrator*
LAB-MGMT01	Create a dynamic group in Entra ID (attribute-based: `department -eq "IT"`)
LAB-MGMT01	Enable *Self-Service Password Reset* (SSPR) for the IT department
LAB-W11-01	Test SSPR as TestUser01 via `aka.ms/sspr`
LAB-MGMT01	Configure **Privileged Identity Management (PIM)**: make the Global Administrator role *eligible* for labadmin
LAB-MGMT01	Activate the GA role Just-in-Time via PIM → verify the audit trail under **Entra** → **PIM** → **Audit history**

Lab commands

```
# Bulk-create on-premises AD users from CSV
Import-Csv .\users.csv | ForEach-Object {
    New-ADUser -Name $_.DisplayName -UserPrincipalName $_.UPN -AccountPassword
(ConvertTo-SecureString $_.Password -AsPlainText -Force) -Enabled $true
}

# Create a dynamic group via Microsoft Graph PowerShell
Connect-MgGraph -Scopes "Group.ReadWrite.All"
New-MgGroup -DisplayName "IT Department" -MailEnabled:$false -
SecurityEnabled:$true `
    -GroupTypes @("DynamicMembership") `
    -MembershipRule '(user.department -eq "IT")' `
    -MembershipRuleProcessingState "On" -MailNickname "it-dept"

# List PIM eligible role assignments
Get-MgRoleManagementDirectoryRoleEligibilitySchedule -All | Select-Object
PrincipalId, RoleDefinitionId, Status
```

Knowledge check

1. What is the principle of *least privilege* when assigning admin roles?
2. What is the difference between a security group and a Microsoft 365 group?
3. How does *bulk user creation* work via the Microsoft 365 admin center?
4. When do you use *dynamic groups* versus *assigned groups*?
5. What is *PIM* and why is *Just-in-Time* access preferred over permanent role assignment?

▶ Answers

Week 3 — Entra ID and hybrid identity

> **Exam domain:** Implement and manage Microsoft Entra identity and access · **Weight:** 25–30%

> **Real-world scenario:** A Sogeti consultant is working with a manufacturing company that has two separate Active Directory forests following a recent acquisition. The acquired subsidiary (300 users) needs to be integrated into the parent company's Microsoft 365 tenant without deploying additional on-premises infrastructure. The parent company already uses Entra Connect Sync for its primary forest. The consultant must evaluate whether Entra Cloud Sync can cover the subsidiary's forest and configure MFA and B2B access for cross-subsidiary collaboration.

Learning Objectives

- ☐ Compare Microsoft Entra Connect Sync and Entra Cloud Sync and select the appropriate solution
- ☐ Configure and verify Multi-Factor Authentication (MFA) using Entra ID security settings
- ☐ Distinguish between per-user MFA, Security Defaults, and Conditional Access-based MFA
- ☐ Invite and manage external B2B guest users and configure cross-tenant access settings
- ☐ Use Entra Connect Health to monitor synchronisation status and investigate errors
- ☐ Configure Entra Password Protection with a custom banned password list

MS Learn modules

- Manage identity synchronisation with Azure AD Connect and Entra Cloud Sync
- Implement multifactor authentication
- Manage external identities

Key Concepts

Term	Description
Entra Connect Sync	Full-featured on-premises synchronisation engine; supports complex topologies, custom attribute mapping, and filtering rules
Entra Cloud Sync	Lightweight cloud-provisioning agent requiring minimal on-premises footprint; ideal for disconnected forests or simple scenarios
Security Defaults	Preconfigured baseline security settings enabled by default on new tenants; requires MFA for all users and blocks legacy authentication
Per-user MFA	Legacy method of enabling MFA individually per account; bypasses Conditional Access and is not recommended for new deployments

Term	Description
Seamless SSO	Feature of Entra Connect that silently authenticates domain-joined devices using Kerberos, providing single sign-on to cloud apps without additional prompts
Azure AD B2B	External identity model allowing guest users from partner organisations to access resources in your tenant using their own credentials
Azure AD B2C	Consumer identity platform for building customer-facing applications with custom sign-up/sign-in flows; separate service from B2B
Entra Password Protection	Service that enforces a global and custom banned password list both in Entra ID and on-premises AD via a proxy agent
Entra Connect Health	Monitoring service that provides health alerts, performance data, and usage analytics for Entra Connect Sync agents

Lab exercises (SSW-Lab)

VM	Task
LAB-DC01	Check sync status: `Get-ADSyncScheduler` → verify cycle
LAB-MGMT01	Configure MFA via **Entra admin center** → **Security** → **Multifactor authentication**
LAB-W11-01	Register MFA method as TestUser01 (Authenticator app)
LAB-MGMT01	Review *Sign-in logs* in Entra ID → filter on MFA events
LAB-MGMT01	Invite an external user (B2B guest) via Entra ID
LAB-MGMT01	Configure *Cross-tenant access settings* for external organisations
LAB-MGMT01	Review **Entra Connect Health** → verify sync agent status and error report
LAB-MGMT01	Enable **Entra Password Protection** → configure banned password list

Lab commands

```
# Check Entra Connect sync scheduler status
Get-ADSyncScheduler

# Trigger a manual delta sync
Start-ADSyncSyncCycle -PolicyType Delta

# Invite a B2B guest user via Microsoft Graph
Connect-MgGraph -Scopes "User.Invite.All"
New-MgInvitation -InvitedUserEmailAddress "partner@externalcompany.com" `
    -InviteRedirectUrl "https://myapps.microsoft.com" -
SendInvitationMessage:$true

# Get sign-in log entries filtered to MFA events (requires AuditLog.Read.All)
Get-MgAuditLogSignIn -Filter "authenticationRequirement eq
'multiFactorAuthentication'" -Top 20 |
    Select-Object UserDisplayName, CreatedDateTime, Status
```

Knowledge check

1. What is the difference between per-user MFA and *Security Defaults*?
2. How does *Seamless Single Sign-On* (SSO) work with Azure AD Connect?
3. What is Azure AD B2B and when do you use B2B versus B2C?
4. How do you troubleshoot a sync error in Azure AD Connect?
5. What is the difference between *Microsoft Entra Connect Sync* and *Entra Cloud Sync*?
6. What does *Entra Password Protection* do and how does the on-premises agent work?

▶ Answers

Week 4 — Exchange Online management

> **Exam domain:** Deploy and manage a Microsoft 365 tenant · **Weight:** 25–30%

> **Real-world scenario:** A Sogeti consultant is called in after a mid-sized law firm (180 users) reports that outbound emails are being rejected as spam by external partners. Investigation reveals the firm has no SPF record, DKIM is disabled, and DMARC is absent. Additionally, the firm's shared `info@` mailbox is being used by five staff members simultaneously with no access controls. The consultant must implement email authentication and restructure the mailbox access model.

Learning Objectives

- ☐ Create and manage Exchange Online recipient types: shared mailboxes, room mailboxes, distribution lists, and Microsoft 365 groups
- ☐ Configure mail flow rules (transport rules) to enforce organisational messaging policies
- ☐ Configure anti-spam and anti-phishing policies in Microsoft Defender for Office 365
- ☐ Perform a message trace to investigate mail delivery issues
- ☐ Configure DKIM signing for a custom domain and understand its relationship with SPF and DMARC

MS Learn modules

- Manage Exchange Online recipients and permissions
- Manage Exchange Online mail flow
- Manage Exchange Online protection

Key Concepts

Term	Description
Shared mailbox	A mailbox accessed by multiple users without a dedicated license (up to 50 GB); no dedicated sign-in credentials
Room mailbox	A resource mailbox representing a physical meeting room; accepts or declines meeting requests automatically
Distribution list	A mail-enabled group that delivers messages to all members; does not provide collaboration features
Mail flow rule (transport rule)	Server-side rule applied to messages in transit; can modify, redirect, reject, or add disclaimers to messages
DKIM	DomainKeys Identified Mail — adds a cryptographic signature to outbound messages, allowing receivers to verify the message was not tampered with
SPF	Sender Policy Framework — a DNS TXT record listing IP addresses authorised to send email for a domain
DMARC	Domain-based Message Authentication, Reporting and Conformance — policy governing what receivers do when SPF or DKIM checks fail (none / quarantine / reject)
Message trace	Exchange Online tool that tracks the path and delivery status of individual messages; available in EAC and via PowerShell

Lab exercises (SSW-Lab)

VM	Task
LAB-MGMT01	Create shared mailboxes via the Exchange Admin Center (EAC)

VM	Task
LAB-MGMT01	Configure a *Distribution list* and a *Microsoft 365 Group*
LAB-MGMT01	Set up a *mail flow rule*: append disclaimer to outbound mail
LAB-MGMT01	Configure *Anti-spam* and *Anti-phishing* policies in Defender for Office 365
LAB-W11-01	Run a *message trace* via EAC → analyse delivery status
LAB-MGMT01	Configure *DKIM* and review DMARC settings for the tenant domain

Lab commands

```
# Connect to Exchange Online PowerShell
Connect-ExchangeOnline -UserPrincipalName admin@ssw.lab

# Create a shared mailbox
New-Mailbox -Shared -Name "Info" -DisplayName "Info Shared Mailbox" -Alias "info"

# Grant Full Access and Send As permissions
Add-MailboxPermission -Identity "info" -User "user1@ssw.lab" -AccessRights
FullAccess -InheritanceType All
Add-RecipientPermission -Identity "info" -Trustee "user1@ssw.lab" -AccessRights
SendAs -Confirm:$false

# Run a message trace for the last 48 hours
Get-MessageTrace -SenderAddress "sender@external.com" -StartDate (Get-
Date).AddHours(-48) -EndDate (Get-Date) |
    Select-Object Received, SenderAddress, RecipientAddress, Subject, Status
```

Knowledge check

1. What is the difference between a *shared mailbox* and a *room mailbox*?
2. How does *message trace* work and when do you use it?
3. What do *Safe Attachments* and *Safe Links* do in Defender for Office 365?
4. What is the difference between *anti-spam* and *anti-phishing* policies?

▶ Answers

Week 5 — SharePoint Online and Microsoft Teams

> **Exam domain:** Deploy and manage a Microsoft 365 tenant · **Weight:** 25–30%

> **Real-world scenario:** A Sogeti consultant is advising a consulting firm (500 users, multiple client engagement teams) that is restructuring its Microsoft 365 collaboration environment. Client teams complain that external sharing is too open — anyone with a link can access project files. Meanwhile, the compliance officer wants sensitivity labels applied to all client-facing Teams and SharePoint sites to enforce data classification. The consultant must tighten external sharing controls and implement a labelling structure.

Learning Objectives

- ☐ Create and configure SharePoint site collections and manage permissions at site, library, and item level
- ☐ Configure external sharing settings in SharePoint and understand the relationship with Entra B2B
- ☐ Create and manage Microsoft Teams teams via the Teams admin center
- ☐ Configure Teams meeting policies and restrict specific capabilities for guests
- ☐ Apply sensitivity labels to Teams and SharePoint sites to enforce data protection settings
- ☐ Use PowerShell (PnP module and Teams module) for bulk management tasks

MS Learn modules

- Manage SharePoint Online
- Manage Microsoft Teams
- Manage Teams collaboration settings

Key Concepts

Term	Description
Team site (Group site)	SharePoint site connected to a Microsoft 365 group; includes shared mailbox, calendar, and Teams channel
Communication site	Broadcast-style SharePoint site for publishing content to a wide audience; not connected to a Microsoft 365 group
Hub site	SharePoint site that aggregates navigation, search, and branding across multiple associated sites
External sharing levels	Configurable at tenant and site level: Anyone, New and existing guests, Existing guests only, or Only people in your organisation
Teams meeting policy	Admin-defined policy controlling what meeting features are available to users (recording, transcription, lobby, guest access)
Sensitivity label on sites	When a label with site and group protection is applied, it enforces privacy settings, external user access, and unmanaged device restrictions on the container
PnP PowerShell	Community-supported PowerShell module for SharePoint Online, providing cmdlets for bulk site, library, and permission management

Term	Description
Teams PowerShell module	Official Microsoft module for managing Teams settings, policies, and membership at scale

Lab exercises (SSW-Lab)

VM	Task
LAB-MGMT01	Create a SharePoint site collection (Team site) and assign permissions
LAB-MGMT01	Configure *external sharing* settings in the SharePoint admin center
LAB-W11-01	Upload documents to SharePoint → test sharing with TestUser02
LAB-MGMT01	Create a Teams team via the Teams admin center → add members
LAB-MGMT01	Configure *Meetings policies* in Teams: restrict recording for guests
LAB-MGMT01	Review **Teams usage reports** in the M365 admin center

Lab commands

```
# Connect to SharePoint Online via PnP
Connect-PnPOnline -Url "https://sswlab.sharepoint.com" -Interactive

# Set external sharing level on a specific site to existing guests only
Set-PnPTenantSite -Url "https://sswlab.sharepoint.com/sites/ClientProject" -
SharingCapability ExistingExternalUserSharingOnly

# Connect to Microsoft Teams module and list all teams with their guest settings
Connect-MicrosoftTeams
Get-Team | Select-Object DisplayName, AllowGuestCreateUpdateChannels,
AllowGuestDeleteChannels | Export-Csv .\teams-guest-settings.csv -
NoTypeInformation

# Apply a sensitivity label to a Teams team (requires label GUID from Get-Label
in compliance module)
Set-Team -GroupId "<team-group-id>" -Sensitivity "<label-GUID>"
```

Knowledge check

1. What is the difference between a *Group site*, *Communication site* and *Hub site* in SharePoint?
2. How do you manage external access in Teams at the channel level versus the team level?
3. What are *sensitivity labels* and how do you apply them to Teams and SharePoint?
4. How do you use PowerShell (PnP / Teams module) for bulk management?

▶ Answers

Week 6 — Microsoft Defender XDR and threat management

> **Exam domain:** Manage security and threats by using Microsoft Defender XDR · **Weight:** 30–35%

> **Real-world scenario:** A Sogeti consultant is engaged by a retail company (1,200 users) following a ransomware incident. The attacker gained initial access via a phishing email, moved laterally across three endpoints, and encrypted a file server before being detected. The client had no EDR solution in place. The consultant must onboard all Windows devices to Defender for Endpoint, set up an incident response workflow in the Defender XDR portal, and implement Attack Simulation Training to reduce phishing susceptibility across the workforce.

Learning Objectives

- ☐ Navigate the Microsoft Defender XDR unified portal and understand its component coverage
- ☐ Onboard a Windows 11 device to Microsoft Defender for Endpoint via Intune policy
- ☐ Investigate a security incident using the Attack story graph and take containment actions
- ☐ Understand how Automated Investigation and Response (AIR) processes alerts
- ☐ Use Threat Explorer to analyse email-based threats in the tenant
- ☐ Interpret the Microsoft Secure Score and implement a scored improvement action
- ☐ Explain the relationship between Exposure Management and Secure Score

MS Learn modules

- Explore the Microsoft Defender XDR portal
- Manage Microsoft Defender for Office 365
- Manage Microsoft Secure Score and Exposure Management

Key Concepts

Term	Description
Microsoft Defender XDR	Extended Detection and Response platform unifying MDE, MDO, MDI, and MDCA signals into correlated incidents
Defender for Endpoint (MDE)	Endpoint security solution providing next-generation protection, EDR, Advanced Hunting, and vulnerability management
Defender for Office 365 Plan 1	Adds Safe Attachments, Safe Links, and anti-phishing policies to Exchange Online Protection
Defender for Office 365 Plan 2	Adds Threat Explorer, Attack Simulation Training, Automated Investigation and Response, and Advanced Hunting on top of Plan 1

Term	Description
Automated Investigation and Response (AIR)	Automated workflow that investigates alerts, collects evidence, and proposes or executes remediation actions
Threat Explorer	Real-time interface for investigating email threats across the tenant; shows delivery actions, detection technology, and message details
Microsoft Secure Score	Aggregated security posture score based on enabled controls across identity, devices, apps, and data
Exposure Management	Broader attack surface management view that combines device vulnerability data, identity risk, and lateral movement paths
Attack Simulation Training	Managed phishing simulation service that measures user susceptibility and assigns targeted awareness training

Lab exercises (SSW-Lab)

VM	Task
LAB-MGMT01	Open the **Microsoft Defender XDR portal** (security.microsoft.com)
LAB-W11-01	Onboard W11-01 to Defender for Endpoint via Intune policy
LAB-W11-01	Simulate suspicious activity: download the EICAR test file → check alert
LAB-MGMT01	Analyse the incident in the Defender portal → review the *Attack story* graph
LAB-MGMT01	Review the **Exposure Management** dashboard → check the Microsoft Secure Score
LAB-MGMT01	Run *Attack Simulation Training* → phishing simulation to TestUser01
LAB-MGMT01	Analyse *Secure Score* → select an improvement action and implement it

Lab commands

```
# Advanced Hunting query — find EICAR-related alerts on onboarded devices
# Run in Defender XDR portal > Advanced Hunting (KQL)
# DeviceAlertEvents
# | where Title contains "EICAR"
# | project Timestamp, DeviceName, Title, Severity, AlertId

# Isolate a device via Microsoft Graph Security API (requires
SecurityEvents.ReadWrite.All)
Connect-MgGraph -Scopes "SecurityEvents.ReadWrite.All"
Invoke-MgRestMethod -Method POST `
    -Uri "https://graph.microsoft.com/v1.0/security/alerts/<alert-id>/comments" `
    -Body (@{ comment = "Isolating device pending investigation" } | ConvertTo-
Json)

# Get Secure Score via Microsoft Graph
Connect-MgGraph -Scopes "SecurityEvents.Read.All"
Get-MgSecuritySecureScore -Top 1 | Select-Object CurrentScore, MaxScore,
CreatedDateTime
```

Knowledge check

1. What is the difference between Defender for Office 365 Plan 1 and Plan 2?
2. How does *Automated Investigation and Response* (AIR) work in Defender?
3. What does *Threat Explorer* show and when do you use it?
4. How do you effectively improve your Microsoft Secure Score?
5. What is *Exposure Management* and how does it relate to Secure Score?

▶ Answers

Week 7 — Microsoft Purview Compliance

> **Exam domain:** Manage compliance by using Microsoft Purview · **Weight:** 10–15%

> **Real-world scenario:** A Sogeti consultant is working with a financial services firm (600 users) that must comply with MiFID II communications monitoring requirements. The firm's legal team has also received a litigation hold request for all emails involving a specific former employee. The compliance officer needs to implement Communication Compliance policies for regulated staff, place an eDiscovery hold on the former employee's mailbox, and configure DLP policies to prevent inadvertent sharing of client financial data.

Learning Objectives

- ☐ Create and publish sensitivity labels with encryption and content marking
- ☐ Configure a DLP policy targeting sensitive information types across Exchange, SharePoint, and OneDrive
- ☐ Test and verify DLP policy enforcement in a controlled scenario
- ☐ Distinguish between sensitivity labels and retention labels and explain when to use each
- ☐ Run a Core eDiscovery search on a user mailbox and export the results
- ☐ Explain the role of Communication Compliance and the scenarios in which it is required

MS Learn modules

- Implement Microsoft Purview Information Protection
- Implement data loss prevention
- Manage Microsoft Purview eDiscovery

Key Concepts

Term	Description
Sensitivity label	A persistent classification tag applied to files, emails, or containers (sites/teams) that can enforce encryption, content marking, and access restrictions
Retention label	A label applied to a specific item (email, document, Teams message) that governs how long the item must be kept and what happens after the retention period
Retention policy	A location-scoped policy that automatically applies a retention or deletion rule to all content in a defined location (mailbox, site, Teams channel)
DLP policy	A Purview rule set that detects sensitive information types (SSNs, credit card numbers, etc.) in content and takes configured actions (block, warn, notify)
Sensitive Information Type (SIT)	A pattern definition (regular expression + keyword list) used by DLP policies and auto-labelling to identify specific categories of sensitive data
eDiscovery (Standard)	Provides search and hold capabilities across Exchange, SharePoint, and Teams for legal investigation purposes
eDiscovery (Premium)	Extends Standard with custodian management, review sets, AI-assisted analysis, near-duplicate detection, and email threading
Communication Compliance	Purview feature that scans Teams and email communications for policy violations (harassment, regulatory content, insider risk signals)

Lab exercises (SSW-Lab)

VM	Task
LAB-MGMT01	Open the **Microsoft Purview portal** (compliance.microsoft.com)

VM	Task
LAB-MGMT01	Create a *sensitivity label*: "Confidential - Internal" with encryption
LAB-W11-01	Apply the label to a Word document → verify encryption
LAB-MGMT01	Create a *DLP policy*: block sending of Social Security Numbers via email
LAB-W11-01	Test the DLP policy: send email with a fictitious SSN → verify block
LAB-MGMT01	Run a *Core eDiscovery* search on the TestUser01 mailbox

Lab commands

```
# Connect to Security & Compliance (Purview) PowerShell
Connect-IPPSSession -UserPrincipalName admin@ssw.lab

# Create a retention policy for Exchange mailboxes (retain 5 years)
New-RetentionCompliancePolicy -Name "Exchange-Retain-5Y" -ExchangeLocation All -
RetentionAction Keep -RetentionDuration 1825

# Create an eDiscovery hold on a specific mailbox
New-CaseHoldPolicy -Name "Litigation-Hold-TestUser01" -Case "LitigationCase01" -
ExchangeLocation "testuser01@ssw.lab"

# Search for content in an eDiscovery case
New-ComplianceSearch -Name "TestUser01-MailSearch" -ExchangeLocation
"testuser01@ssw.lab" -ContentMatchQuery "subject:confidential"
Start-ComplianceSearch -Identity "TestUser01-MailSearch"
```

Knowledge check

1. What is the difference between *sensitivity labels* and *retention labels*?
2. How does *DLP endpoint protection* work on managed devices?
3. What is *Communication Compliance* and when is it required?
4. What is the difference between *Core eDiscovery* and *eDiscovery Premium*?

▶ Answers

Week 8 — Exam preparation

Activities

- Review weak domains based on the official exam study guide
- Complete the **Microsoft Learn practice assessment** for MS-102: Practice assessment
- Revisit Azure AD Connect sync, Defender incidents and Purview labels

- Practice PowerShell: MgGraph module (`Connect-MgGraph`), Exchange Online module
- Schedule your exam via Pearson VUE

Exam focus areas

- Hybrid identity: know both **Connect Sync** and **Entra Cloud Sync** — differences and migration scenarios
- **Defender XDR (30–35%):** unified portal, AIR, Exposure Management, Attack Simulation, Secure Score improvements
- DLP: differences between policies for Exchange, SharePoint and Endpoint
- **Purview (10–15%):** retention vs. sensitivity labels — know core scenarios
- PIM: eligible vs. active vs. permanent role assignments in MS-102 scope
- Entra Password Protection and Connect Health are now explicit exam objectives

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Exam Coverage Gaps and Must-Do Labs

This section addresses topics that are explicitly tested on the MS-102 exam but are underrepresented in the eight-week programme above. Complete these before scheduling the exam.

Topics still requiring attention

1. **Service Health and operational monitoring** — The exam tests knowledge of how to monitor tenant health, configure service health notifications, and use the Network connectivity dashboard. The Message Center workflow (filtering, tagging, and exporting announcements) is also in scope.

2. **Group-based licensing** — Assigning licenses via group membership rather than per-user is an exam objective. You must understand how group-based licensing interacts with license conflicts, error states, and the processing delay after group membership changes.

3. **Microsoft Defender for Cloud Apps (MDCA)** — Cloud Discovery, app sanctioning/unsanctioning, activity policies, session policies, and the Conditional Access App Control (CAAC) proxy are all within Defender XDR domain scope (30–35%). These topics are absent from the weekly programme.

4. **Purview retention policies and retention labels (side-by-side)** — The exam distinguishes carefully between location-scoped retention policies and item-scoped retention labels. Records management concepts (record declaration, disposition review, file plan) are also in scope for Domain 4.

5. **Defender for Identity (MDI) fundamentals** — MDI sensor installation, alert types (Pass-the-Hash, Kerberoasting, lateral movement paths), and how MDI integrates with Defender XDR incidents are tested and not covered in the weekly labs.

6. **Conditional Access policy configuration** — CA is explicitly in Domain 2 (Entra identity and access). Named locations, device compliance conditions, authentication strengths, and sign-in frequency controls are common exam scenarios.

Must-do labs before exam day

1. **Service Health notifications:** In the M365 admin center, configure email notifications for service incidents affecting Exchange Online and Teams. Document which advisory and incident categories are available and what triggers a notification.

2. **Group-based licensing:** Create a security group in Entra ID, assign an E5 license to the group, add TestUser01 as a member, and verify that the license is applied automatically. Then remove the user from the group and verify the license is revoked. Observe the processing delay.

3. **Defender for Cloud Apps — Cloud Discovery:** In the Defender XDR portal, navigate to Cloud apps → Cloud Discovery. Review the top discovered apps, sanction one app and unsanction another, and observe how unsanctioned app blocking works via Defender for Endpoint integration.

4. **Retention policy + retention label comparison:** Create one Exchange-scoped retention policy (retain for 5 years, then delete) and one retention label that declares items as records (retain for 7 years, disposition review required). Apply both in your lab tenant and compare the user experience and admin audit trail.

5. **Three complete Defender XDR incident flows:** For each incident, practice the full workflow: triage the alert queue, open the incident, review the Attack story, identify affected entities (user, device, email), execute a containment action (isolate device or disable user), and close the incident with a classification and determination. Use EICAR and Attack Simulation to generate real incidents in the lab.

6. **Conditional Access — named location and compliance policy:** Create a CA policy that requires compliant device status for access to SharePoint Online. Test from an enrolled (compliant) device and an unenrolled device to observe the different outcomes. Review the sign-in log entries to understand the CA evaluation detail.

Exit criteria before scheduling the exam

1. You can demonstrate at least two concrete administrative actions for each of the four MS-102 exam domains without referring to notes.
2. You consistently score above the 700/1000 pass mark equivalent on the Microsoft Learn practice assessment.
3. You have completed at least one full incident response flow in the Defender XDR portal from alert through to resolved classification.
4. You can explain the difference between PHS, PTA, and Cloud Sync without prompting, including when to choose each.
5. You have no unresolved gaps in the Purview domain: you can confidently distinguish sensitivity labels, retention labels, retention policies, DLP policies, and eDiscovery (Standard vs. Premium).